

Propagation of LCL Filter Degradation to Distribution Transformer Loss of Life: Analytical Limits and Mission-Profile-Dependent Aging of Grid-Connected Inverters

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Abstract

The harmonic emission of grid-connected inverters is conventionally treated as a property fixed at commissioning and verified once. In reality, the LCL filter components that shape this emission degrade: self-healing in metallized-film capacitors progressively reduces capacitance and increases dielectric loss, shifting the filter resonance and detuning damping designed for nominal parameters. Field evidence has established the endpoints of a damaging chain — converter harmonics above 2 kHz measurably increase distribution transformer eddy losses and accelerate insulation aging, and filter-capacitor drift is real enough to have motivated dedicated condition-monitoring methods — yet the middle link has not been quantified: how multi-year filter degradation reshapes the emitted spectrum and thereby consumes the life of the transformer it feeds. This paper closes that link analytically and numerically. A degradation-resolved model propagates film-capacitor aging trajectories through the delay-limited active-damping dynamics of a digitally controlled inverter to a time-evolving harmonic and interharmonic emission profile; a generalized harmonic loss factor, bounded between the IEEE C57.110 weighting and a skin-effect-corrected exponent, drives an IEC 60076-7 hot-spot model to yield the differential transformer loss of life attributable solely to filter aging. Two closed-form safe limits are derived: a critical capacitance-drift threshold beyond which the resonance enters the delay-induced negative-damping region, and a maximum tolerable resonant-band current — approximately 1.4–2.6% of rated current for representative parameters — beyond which transformer lifetime consumption exceeds a stated budget. Mission profiles with and without harmonic-compensation duty are compared, the former closing a resonance-mediated degradation feedback loop through capacitor self-heating. Results give asset managers a quantitative basis for pricing ancillary-service commitments of aging inverter fleets.

Index Terms — LCL filter, film capacitor degradation, harmonic loss factor, transformer loss of life, active damping, mission profile, reliability, supraharmonics.

A. Real-World Context

Three developments converge to make the question of this paper operationally urgent.

First, the world's first large photovoltaic fleets are aging past the design assumptions of their power-quality verification. Germany alone connected over 20 GW of PV during the 2010–2012 feed-in-tariff boom; those plants are now 14–16 years into service. Their inverters were type-tested for harmonic emission once, at commissioning, against limits in IEEE 519, IEEE 1547, and, in Europe, VDE-AR-N 4105 and related grid codes. No mechanism in routine operation re-verifies that emission as passive components drift. Low-voltage feeders are, in practice, sparsely instrumented for power quality: a slow rise in high-frequency emission can persist for years between audits — a silent erosion of compliance that harms equipment before it triggers any protection.

Second, the same aging inverters are being asked to do more. Grid codes amended over the past decade (IEEE 1547-2018 and its successors; VDE-AR-N 4105/4110) require or enable reactive-power support, and harmonic-compensation duty for multifunctional inverters is an active deployment question. Each added duty changes the electrical and thermal stress on the very filter components whose degradation this paper studies. An asset manager deciding whether to commit a 15-year-old fleet to compensation duty currently has quantitative tools for the inverter-side lifetime cost of that decision, but none for the transformer-side cost.

Third, the frequency range in which the relevant emission lives — above 2 kHz, the supraharmonic range — sits in a recognized standardization gap. Compatibility levels for 2–9 kHz have only recently entered IEC 61000-2-2, normative emission limits are still maturing, and transformer derating practice (IEEE C57.110) was developed and validated for integer harmonics typically up to order 25–50, with a loss-weighting law known to be inaccurate precisely where converter resonances emit. The assets most exposed are distribution transformers: 30–40-year service-life devices, fleet-replaced slowly, whose insulation aging is exponential in hot-spot temperature so that even single-kelvin chronic increments translate into measurable lifetime consumption.

The physical chain at issue is simple to state. The LCL filter of a grid-connected inverter contains a metallized-film capacitor whose capacitance falls and whose dielectric loss rises over years of service. Falling capacitance shifts the filter resonance upward; in digitally controlled inverters the achievable damping at the resonance is delay-limited and frequency-dependent, so the shift simultaneously detunes the damping. The result is a slowly growing current emission concentrated in a band near the (moving) resonant frequency — for the reference design of this paper, near 2.4 kHz, squarely in the under-standardized range. That current flows through the windings of the distribution transformer at the point of common coupling, where eddy losses grow approximately with the square of frequency, raising the winding hot-spot temperature and accelerating the Arrhenius-governed aging of the cellulose insulation. One root cause — filter degradation

— thus produces two coupled end-of-life trajectories: the inverter's own stability margin and the transformer's remaining insulation life.

B. Problem Statement

Every individual link of this chain is established in its own literature, as Section II documents in detail. What does not exist is the propagation: no published work drives a transformer loss-of-life calculation from a *time-evolving* converter degradation model, and no published work provides limits — on capacitance drift or on resonant-band emission — that separate the benign regime from the damaging one. This paper supplies both, in closed form where possible, and quantifies how the answer depends on the inverter's mission profile.

C. Contributions

1. To the authors' knowledge, the first propagation of deterministic LCL degradation trajectories through the time-evolving harmonic and interharmonic emission spectrum to distribution-transformer loss of life — an integration of individually established links not previously coupled into one quantified chain (Sections IV-A through IV-F).
2. The first extension of the mission-profile reliability framework across the point of common coupling: the transformer-side lifetime cost of committing inverters to harmonic-compensation duty, expressed as differential loss of life between profiles (Sections IV-F, IV-H).
3. A system-level, resonance-mediated aging feedback closure — capacitance loss shifts the resonance and detunes the delay-limited active damping, and the growing resonant current through the capacitor branch accelerates its own degradation — extending published component-level self-heating/ESR coupling models through the electrical network (Section IV-H).
4. Closed-form safe limits with a bounded operating-regime map: a critical capacitance-drift threshold $\delta_C^* = 1 - \rho_0^2$ tied to the designer's resonance-placement margin, and a maximum tolerable resonant-band current I_{res}^{max} tied to a transformer lifetime budget, evaluated against existing grid-code emission limits (Section IV-G).

D. Organization

Section II reviews the five bodies of prior work this paper builds on and identifies, for each, what it does not do. Section III condenses the gaps and maps them to the contributions. Section IV develops the analytical framework and derives the safe limits. Section V outlines the numerical validation and sensitivity campaign. Section VI concludes.

II. State of the Art

A. Harmonic Effects on Transformer Thermal Aging

The standardized machinery is mature. IEEE Std C57.110 [1] defines the winding eddy-current harmonic loss factor $F_{HL} = \sum_h h^2 I_h^2 / \sum_h I_h^2$ and the UL-style K-factor $K = \sum_h h^2 I_h (pu)^2$, on the premise that winding eddy loss scales with the square of frequency; other stray losses are weighted with exponent 0.8. IEEE Std C57.91 [2] and IEC 60076-7 [3] supply the thermal models — top-oil and hot-spot dynamics and the aging-acceleration factor F_{AA} , exponential in hot-spot temperature with a doubling per approximately 6 K — that convert losses into insulation loss of life. The European Factor-K formulation [4] uses a reduced harmonic exponent ($q \approx 1.7$ for round/rectangular conductors, 1.5 for foil), reflecting the long-known fact, acknowledged in C57.110 itself, that skin effect makes the h^2 law increasingly conservative at high harmonic order; corrected loss-factor formulations exist [5], and analyses of large-conductor windings report the squared-frequency assumption failing above roughly the fifth harmonic [6].

Application studies are numerous. Distribution-transformer derating and loss-of-life under PV, EV-charger, and general nonlinear loading has been computed in many works [7]–[10]; a field-measurement study of grid-connected inverter emission in the 0–9 kHz range [7] reports high-frequency harmonics increasing transformer eddy losses by roughly a third and accelerating aging by over twenty percent, with components above 2 kHz contributing strongly to the K-factor — direct empirical anchoring of this paper's downstream endpoint. Solar-farm inverter transformers have been life-assessed with measured load, harmonic, moisture, and temperature data driving an extended hot-spot model [11].

What this body of work does not do: in every cited study the harmonic spectrum is an exogenous input — measured at one time, or parameterized by load or irradiance. None models the spectrum as a function of the emitting converter's internal degradation state, and consequently none can compute the transformer-life consequence of converter aging.

B. LCL Resonance, Digital Control, and Stability

LCL filters attenuate switching harmonics economically but introduce a resonance between the filter capacitance and the parallel combination of converter-side and grid-side inductances [12]. In digitally controlled converters the total loop delay — one sampling period of computation plus half a period of modulator hold, $T_d = 1.5T_s$ — fundamentally limits damping. Two canonical results organize the field. For single-loop control, the feedback variable determines an inherent stability region bounded by the critical frequency $f_s/6$: converter-side current feedback is stable without dedicated damping only for $f_{res} < f_s/6$, grid-side current feedback only for $f_{res} > f_s/6$ [13]. For capacitor-current-feedback (CCF) active damping, the delay converts the intended virtual resistor into a frequency-dependent virtual impedance whose resistive part is positive below $f_s/6$ and negative above it [14]; a resonance migrating toward and across $f_s/6$ therefore loses damping continuously and then is actively destabilized. Grid-impedance variation shifts the effective resonance — an experimental design with a 5.2 kHz nominal resonance

measured it near 3.9 kHz purely due to the impedance of the coupling transformer [15] — and weak-grid robustness is an active design topic [16]. Interval and multiparameter-uncertainty analyses of converter small-signal stability explicitly motivate themselves by long-term parameter divergence from nominal due to natural aging [17].

What this body of work does not do: parameter variation is treated as a static tolerance or an uncertainty set. No work assigns parameters a time axis — a degradation trajectory — and follows the stability margin and, crucially, the *emission* along it.

C. Filter-Capacitor Degradation Physics and Condition Monitoring

Metallized polypropylene film capacitors, the standard choice for AC filter duty, wear out by accumulated self-healing breakdowns that remove electrode area: capacitance falls monotonically (end-of-life criteria of 2–5% loss are typical) while dissipation factor and ESR rise, with degradation rate accelerated by temperature, voltage, and especially humidity; manufacturer endurance and temperature-humidity-bias data and lifetime models of Arrhenius/inverse-power form quantify the rates [18], [19]. The coupling between hot-spot temperature and ESR growth — loss begets heat begets loss — has been modeled at component level, including the influence of harmonic content on metallized-film lifetime [20]. The drift is observable enough to support a condition-monitoring literature: capacitance estimation of LCL filter capacitors via injected resonance response [21], and digital-twin parameter estimation of output-filter degradation [22]. A diagnostic patent infers converter aging state from harmonic signatures of the output current [23].

What this body of work does not do: it stops at detection or at the component's own time-to-failure. The downstream electrical consequence of the drift — the reshaped emission — is used at most as a diagnostic signature, never propagated as a stressor to other grid assets.

D. Mission-Profile Reliability of Grid-Connected Converters

The mission-profile method — translating site irradiance, temperature, and duty into component thermal cycling and Arrhenius wear, then into lifetime distributions — is an established framework for PV inverters [24], [25]. Reactive-power duty has been shown to reduce inverter lifetime and reliability in mission-profile studies [26], trade-offs between reactive capability, sizing ratio, and lifetime have been quantified across climates [27], and — closest to the present work — reactive-power injection has been propagated specifically to LCL filter-capacitor time-to-failure, including ESR growth over the operating period, in a 2 MW wind-converter case study [28].

What this body of work does not do: its system boundary is the converter. The lifetime currency is the converter's own components. No mission-profile study crosses the point of common coupling to ask what the duty costs the transformer.

E. Supraharmonics (2–150 kHz)

Emission in the 2–150 kHz range from converter-dense networks is an active power-quality field: measurement methods are being standardized [29], compatibility levels for the lower portion of the band have entered IEC 61000-2-2, and reviews document the asset-side

effects — additional losses and heating of conductors, capacitors, and transformers, and aging of dielectric materials [30], [31]. The harmonic-rejection behavior of grid-tied inverters has been analyzed up to 9 kHz with all internal and external distortion sources represented [32].

What this body of work does not do: supraharmonic emission is characterized as a present-day phenomenon of device populations. Its evolution with device aging, and a transformer-lifetime budget for it, are absent.

F. Synthesis

Reading A-E together: the transformer-aging community treats emission as static; the stability community treats parameters as static or as uncertainty sets without a time axis; the capacitor community stops at the component terminal; the mission-profile community stops at the converter enclosure; the supraharmonics community characterizes today's emission, not its drift. The chain this paper assembles runs orthogonally through all five.

III. Research Gaps and Contributions

G1 — No degradation-driven emission model. Prior transformer-aging studies take I_h as measured or load-parameterized; prior degradation studies do not output spectra. → **C1:** a degradation-resolved emission trajectory $I(f, t)$ and its propagation to transformer loss of life.

G2 — No transformer-side cost of converter duty. Mission-profile reliability ends at the converter boundary. → **C2:** differential transformer loss of life between energy-export and harmonic-compensation profiles.

G3 — No system-level closure of the aging feedback. Component-level ESR/self-heating coupling exists; the electrical-network-mediated loop (capacitance loss → resonance shift → damping detune → resonant current through the capacitor branch → accelerated capacitance loss) is unmodeled. → **C3:** the coupled degradation ODE of Section IV-H.

G4 — No safe limits. Neither a tolerable capacitance-drift bound for stability nor a tolerable resonant-emission bound for transformer life exists in any standard or paper. → **C4:** the closed-form limits δ_C^* and I_{res}^{max} of Section IV-G, benchmarked against IEEE 519 emission limits.

IV. Analytical Framework: From Capacitance Drift to Transformer Loss of Life

Claim inventory. Equations (1)–(5), (7)–(12), and (15) are standard or published physics — degradation laws [18]–[20], the delay-equivalence (4) [14], and circuit and C57.110/C57.91 thermal relations — assembled here as scaffolding; no novelty is claimed for them, and the interharmonic loss integral (9) and differential comparator (12) are claimed only as methodological choices. The analytical contributions of this paper are exactly four boxed

closed-form results plus one criterion: the damping-erosion leverage $\Lambda(\rho_0)$ (5b); the drift budget δ_C^* (6); the lifetime-budget inversion $\Delta\theta_{max}$ (13); the transformer-derived emission limit I_{res}^{max} (14); and the feedback-strength criterion γ (17). Each is dimensionless or closed-form, design-generic, and checkable by a reviewer in one line.

A. System and Degradation Model

Consider a three-phase voltage-source inverter with LCL filter: converter-side inductance L_1 , filter capacitance C_f with series resistance R_c (ESR plus any damping resistor), grid-side inductance L_2 , and grid inductance L_g (dominated at LV by the coupling-transformer leakage). Define $L'_2 = L_2 + L_g$ and the parallel inductance $L_p = L_1 L'_2 / (L_1 + L'_2)$. Digital control runs at sampling frequency f_s with total loop delay $T_d = 1.5 T_s$ (one period computation, half a period modulator hold). Capacitor-current-feedback (CCF) active damping with gain K_{ad} is assumed; the conclusions transfer to single-loop designs via the same critical frequency.

Film-capacitor degradation is parameterized by the fractional capacitance loss $\delta_C(t)$ and dissipation-factor growth:

$$C_f(t) = C_{f0} (1 - \delta_C(t)), \quad \tan \delta(t) = \tan \delta_0 (1 + \delta_R(t)) \quad (1)$$

with the drift rate governed by an Arrhenius-inverse-power law calibrated to manufacturer endurance and temperature-humidity-bias data:

$$\frac{d\delta_C}{dt} = A \exp\left(-\frac{E_a}{k_B T_h}\right) \left(\frac{V_c}{V_0}\right)^n \quad (2)$$

where T_h is the capacitor hot-spot temperature and V_c the applied voltage stress. Slow/median/fast parameter sets (A, E_a, n) form the degradation scenario band. End-of-life is taken at $\delta_C^{EOL} = 5\%$ per common film-capacitor practice.

B. Resonance Trajectory

The LCL resonance is

$$f_{res}(t) = \frac{1}{2\pi} \sqrt{\frac{L_1 + L'_2}{L_1 L'_2 C_f(t)}} = \frac{f_{res,0}}{\sqrt{1 - \delta_C(t)}} \approx f_{res,0} \left(1 + \frac{1}{2} \delta_C(t)\right) \quad (3)$$

so the logarithmic sensitivity is exactly $-\frac{1}{2}$: a 5% capacitance loss raises the resonant frequency by $\approx 2.5\%$. For the reference design ($f_{res,0} = 2373$ Hz), end-of-life drift places the resonance near 2434 Hz. Equation (3) also makes the inductances' role explicit: any change in L_g (grid/transformer impedance) enters identically, which is why Section V treats grid impedance as a $\pm 30\%$ uncertainty band rather than a deterministic aging channel.

C. Stability Erosion and the Critical Drift Limit

With loop delay $T_d = 1.5T_s$, CCF active damping no longer synthesizes a pure virtual resistor across C_f ; the known result [14] is a frequency-dependent virtual impedance whose resistive part is

$$R_v(\omega) = R_{v0} \cos(1.5 \omega T_s), \quad R_{v0} = \frac{L_1}{C_f K_{ad} K_{pwm}} \quad (4)$$

which is positive for $f < f_s/6$, zero at $f_s/6$, and negative above — the delay-induced negative-damping region. Two quantitative consequences follow.

(i) **Margin erosion.** The total damping resistance in the capacitor branch at the (moving) resonance is $R_d(t) = R_c(t) + \alpha R_v(2\pi f_{res}(t))$ (with α the topology-dependent mapping of the parallel virtual impedance into the series branch), and the resonance quality factor — which sets the amplification of any excitation in the resonant band — is

$$Q(t) = \frac{1}{R_d(t)} \sqrt{\frac{L_p}{C_f(t)}} = \frac{Z_0}{R_d(t) \sqrt{1 - \delta_C(t)}}, \quad Z_0 = \sqrt{L_p / C_{f0}}. \quad (5)$$

The cosine in (4) is steep where designs typically sit. Differentiating (4) along the trajectory (3) and evaluating at $\delta_C \rightarrow 0$ gives the **damping-erosion leverage factor** — the relative damping lost per unit capacitance lost:

$$\Lambda(\rho_0) \equiv -\frac{d \ln R_v}{d \delta_C} = \frac{\pi \rho_0}{4} \tan \frac{\pi \rho_0}{2} \quad (5b)$$

a function of the single dimensionless design parameter $\rho_0 = f_{res,0}/(f_s/6)$. It diverges as $\rho_0 \rightarrow 1$: designs that spend their resonance-placement margin lose damping catastrophically fast per unit drift, while conservative placements ($\rho_0 \leq 0.5, \Lambda \leq 0.4$) are nearly insensitive. For the reference design ($\rho_0 = 0.890, \Lambda = 4.0$), a 5% capacitance loss erodes the virtual resistance by $\approx 21\%$ — four times the relative capacitance change, and eight times the relative frequency shift — the leverage that makes a small component drift electrically visible.

(ii) **The critical drift limit.** Define the designer's resonance-placement margin $\rho_0 = f_{res,0}/(f_s/6) \leq 1$. The capacitance loss at which the resonance reaches the critical frequency follows from (3):

$$\delta_C^* = 1 - \rho_0^2 \quad (6)$$

This single expression converts a control-design choice into a component-aging budget. For the reference design $\rho_0 = 0.890$, giving $\delta_C^* = 20.8\%$ — far beyond film-capacitor end-of-life, so outright instability via boundary crossing is *not* the realistic failure mode here; the operative mechanism is the margin and emission growth of (5). But the budget collapses quadratically as designs use their headroom: $\rho_0 = 0.97$ gives $\delta_C^* = 5.9\%$, inside the end-of-life drift of the capacitor itself. Equation (6) therefore separates inverter designs into a *drift-tolerant* class ($\delta_C^* \gg \delta_C^{EOL}$) and a *drift-critical* class ($\delta_C^* \lesssim \delta_C^{EOL}$) — the first safe limit of this paper. Notch-filter-based damping tightens the budget further, since detuning is governed by the notch bandwidth rather than by (6); this is quantified in Section V.

D. Emission Trajectory

The grid-side current spectrum is driven by two exogenous excitations through degradation-dependent transfer functions:

$$I_2(f, t) = |Y_c(f, t)| V_{inv}(f) + |Y_g(f, t)| V_g(f) \quad (7)$$

where $V_{inv}(f)$ collects PWM sidebands (double-Fourier), dead-time components, and reference distortion, and $V_g(f)$ is background grid distortion at IEC 61000-2-2 / EN 50160 compatibility levels. Near the resonance both admittances are proportional to $Q(t)$, so the band-limited resonant emission

$$I_{res}(t) = \sqrt{\int_{f_{res} \pm B/2} I_2^2(f, t) df} \propto Q(t) E(f_{res}(t)) \quad (8)$$

with $E(\cdot)$ the excitation density sampled *at the moving resonant frequency* — a second drift mechanism, since the resonance migrates across a non-flat excitation landscape. Equations (5) and (8) are the paper's middle link: capacitance drift in, emission trajectory out.

E. Generalized Harmonic Loss Factor and Hot-Spot Sensitivity

Because f_{res} is not an integer multiple of the fundamental (order $h_{res} = f_{res}/f_1 \approx 47.5$ at 50 Hz), the discrete loss factor is generalized to a spectral integral with weighting exponent q :

$$F_{HL}(t) = \frac{\int (f/f_1)^q I_2^2(f, t) df}{\int I_2^2(f, t) df}, \quad q \in [1.6, 2.0] \quad (9)$$

valuated throughout as a band: $q = 2$ reproduces IEEE C57.110 and is a documented conservative ceiling at high order [1], [5], [6]; $q \approx 1.7$ follows the IEC Factor-K practice for

round/rectangular conductors [4]. For a resonant band small against the fundamental ($I_{res} \ll I_1$), the degradation increment is

$$\Delta F_{HL}(t) \approx h_{res}^q \left(\frac{I_{res}(t)}{I_1} \right)^2. \quad (10)$$

The weight h_{res}^q is enormous — 708 at $q = 1.7$, 2256 at $q = 2$ for $h_{res} = 47.5$ — which is precisely why sub-percent currents at the resonance matter thermally.

Per IEEE C57.110 / C57.91 the rated hot-spot rise over top oil scales as $\Delta\theta_{HS} = \Delta\theta_{HS,R} [K_L^2 (1 + F_{HL} P_{EC-R}) / (1 + P_{EC-R})]^m$ with load factor K_L , rated winding-eddy loss ratio P_{EC-R} , and cooling exponent $m \approx 0.8$. The first-order hot-spot sensitivity to the harmonic increment, at rated load and $F_{HL} \approx 1$, is

$$\frac{\partial(\Delta\theta_{HS})}{\partial F_{HL}} = \frac{m \Delta\theta_{HS,R} P_{EC-R}}{1 + P_{EC-R}} \Rightarrow \Delta\theta_h(t) \approx \frac{m \Delta\theta_{HS,R} P_{EC-R}}{1 + P_{EC-R}} h_{res}^q \left(\frac{I_{res}(t)}{I_1} \right)^2 \quad (11)$$

F. Aging Acceleration and Differential Loss of Life

The aging-acceleration factor (thermally upgraded paper, $\theta_{ref}=110^\circ\text{C}$) is $F_{AA} = \exp\left[\frac{15000}{383} - \frac{15000}{\theta_{HS}+273}\right]$, doubling per $\approx 6-7$ K near reference; the IEC relative-aging form $V = 2^{(\theta_{HS}-98)/6}$ is used in parallel for non-upgraded paper. Loss of life over horizon T is $LOL(T) = \frac{1}{L_n} \int_0^T F_{AA} dt$ with $L_n = 180,000$ h. The quantity this paper reports is the *differential* loss of life — the aging-filter case minus a frozen-parameter baseline under identical load and ambient:

$$\Delta LOL(T) = \frac{1}{L_n} \int_0^T F_{AA}^{base}(\tau) \left[2^{\Delta\theta_h(\tau)/6} - 1 \right] d\tau \quad (12)$$

which isolates the filter-aging contribution from the load- and climate-dominated absolute aging, and is the correct comparator between mission profiles.

G. Safe Limits

Tolerable chronic hot-spot increment. Let the lifetime budget be: filter-aging may consume at most an extra fraction ε of insulation life over horizon T , given a baseline aging rate $\bar{F}_{AA}$ during loaded hours. Inverting (12) for a sustained increment:

$$\Delta\theta_{max} = 6 \log_2 \left(1 + \frac{\varepsilon L_n}{\bar{F}_{AA} T} \right) \quad (13)$$

For $\varepsilon = 1\%$, $T = 10$ y, $\bar{F}_{AA} = 0.2$ (a typically loaded distribution unit): $\Delta\theta_{max} \approx 0.85$ K. The exponential aging law is what makes sub-kelvin chronic increments budget-relevant.

Maximum tolerable resonant-band current. Substituting (13) into (11) and solving for the emission:

$$\frac{I_{res}^{max}}{I_1} = \sqrt{\frac{\Delta\theta_{max} (1 + P_{EC-R})}{m \Delta\theta_{HS,R} P_{EC-R} h_{res}^q}} \quad (14)$$

Worked evaluation ($\Delta\theta_{HS,R} = 25$ K, $m = 0.8$, $P_{EC-R} = 0.10$, $h_{res} = 47.5$, $\Delta\theta_{max} = 0.85$ K):

Quantity	$q = 1.7$	$q = 2.0$
h_{res}^q	708	2256
I_{res}^{max} / I_1	2.6%	1.4%

Three readings of this number. First, against grid codes: IEEE 519 limits individual harmonics of order ≥ 35 to 0.3% of demand current (for the stiffest short-circuit-ratio class) — a factor 4.8–8.6 below the transformer-thermal limit. A *compliant* inverter is therefore transformer-safe under this budget; the hazard is confined to the degraded, unmonitored regime in which emission has silently grown past compliance — exactly the regime Section I argued is unobserved on LV feeders. Second, against the eddy-loss ratio: at $P_{EC-R} = 0.20$ the limit tightens to 1.1–1.9%; high-eddy-loss transformer designs are the exposed class. Third, against load: at $K_L < 1$ the limit loosens with $1/K_L^{(\text{via } \Delta\theta_{HS})}$, so lightly loaded transformers tolerate proportionally more. Equations (6) and (14) together constitute the safe-operating envelope: a *component* budget for the inverter designer and an *emission* budget for the asset owner.

H. Resonance-Mediated Degradation Feedback

The resonant current of (8) flows through the capacitor branch. Its self-heating contribution closes a loop that exists nowhere else in the system:

$$T_h(t) = T_a + R_{th} [P_{c,base} + ESR(t) I_{c,res}^2(t)], \quad I_{c,res}(t) \propto Q(\delta_C(t)) E(f_{res}(t))$$

which, inserted into (2), yields the autonomous degradation dynamics

$$\frac{d\delta_C}{dt} = A \exp\left(-\frac{E_a}{k_B [T_a + R_{th} (P_{c,base} + ESR(\delta_C) \kappa^2 Q^2(\delta_C) E^2)]}\right) \left(\frac{V_c}{V_0}\right)^n. \quad (16)$$

Since $dQ/d\delta_C > 0$ for any design whose resonance approaches $f_s/6$ from below (Eqs. 4–5b) and $dESR/d\delta_C > 0$ empirically, the right-hand side is non-decreasing in δ_C — but

whether the loop *materially* accelerates degradation is a quantitative question, answered by the dimensionless loop strength

$$\gamma(t) = \frac{E_a}{k_B T_h^2} R_{th} ESR(t) I_{c,res}^2(t) \quad (17)$$

the resonant self-heating increment expressed in units of the Arrhenius e-folding temperature $k_B T_h^2 / E_a$ (≈ 7 - 11 K for film capacitors); the degradation rate is amplified by e^γ . For a healthy, grid-code-compliant emission level, $\gamma \ll 1$ and the loop is thermally negligible; it activates only in the degraded high- Q regime where $I_{c,res}$ has grown. The feedback therefore does not create the hazard — it sharpens its endgame, and (17) states exactly when. Mission Profile B (harmonic compensation) enters through $P_{c,base}$ — commanded compensation currents raise the capacitor's baseline RMS loading from day one — so the two profiles start the same ODE from different effective stress levels and diverge thereafter. This is the system-level closure of Contribution 3; the component-level couplings it builds on are established [20], while the network-mediated gain $Q(\delta_C)$ and the criterion (17) are not.

V. Validation and Sensitivity Plan

The analytical chain (1)-(16) is verified in three stages. (i) *Electrical:* the small-signal emission trajectory of (7)-(8) is cross-validated against switched time-domain simulation at selected aging snapshots, including the interharmonic structure of the resonant band. (ii) *Thermal:* the linearized sensitivity (11) is checked against the full nonlinear IEC 60076-7 dynamic model driven by a representative annual load/ambient profile for a German LV distribution transformer. (iii) *Statistical:* a global sensitivity analysis (Sobol indices) spans the degradation-rate band of (2), $P_{EC-R} \in [0.05, 0.20]$, $q \in [1.6, 2.0]$, load factor, and $\pm 30\%$ grid impedance — the last covering transformer-impedance uncertainty, including discrete post-through-fault leakage change, in place of any deterministic transformer-to-inverter aging coupling, which Section IV-B showed enters only through L_g and for which no smooth physical driver exists. Headline outputs: threshold-crossing-year distributions per profile against (6); ΔLOL bands per profile against (12)-(14); and the operating-regime map of Contribution 4.

VI. Conclusion

A single root cause — film-capacitor degradation in the LCL filter — has been propagated analytically to two coupled end-of-life trajectories: the inverter's delay-limited damping margin and the distribution transformer's insulation life. The chain is built entirely from individually validated physics, yet produces results that exist nowhere in the literature: the drift budget $\delta_C^* = 1 - \rho_0^2$, which classifies inverter designs into drift-tolerant and drift-